

Spontaneous remission of skull Langerhans cell histiocytosis that had developed by repeated head injury: illustrative case

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BACKGROUND Langerhans cell histiocytosis (LCH) was previously characterized as the proliferation of Langerhans-type histiocytes with a wide range of clinical presentations that arise mostly in children. The typical presentation is a gradually enlarging, painless skull mass. Rapid clinical deterioration is rare.

OBSERVATIONS A 3-year-old boy who had incurred a right frontal impact head injury demonstrated no apparent neurological deficits. He subsequently bruised the same region multiple times. The right frontal swelling gradually increased over the course of 6 days after the initial injury. Skull radiography showed no bony lesion. The same site enlarged markedly 12 days after the initial injury. Magnetic resonance imaging revealed a frontal bony tumorous lesion associated with multiple subcutaneous cystic mass lesions. The patient underwent open biopsy of the skull lesion and evacuation of the subcutaneous lesions. Histopathological examination confirmed the diagnosis of LCH. Immunohistochemical evaluation revealed positivity for CD1a and langerin and no immunopositivity for *BRAF V600E*. The skull lesion spontaneously disappeared 30 days after the biopsy without recurrence.

LESSONS Physicians should be aware of this rare clinical manifestation of LCH that developed by a repeat head injury.

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KEYWORDS skull Langerhans cell histiocytosis; head injury; spontaneous remission; local inflammatory response; inflammatory cytokine

Langerhans cell histiocytosis (LCH) is an uncommon collection of disorders distinguished by an excessive proliferation of histiocytes of Langerhans cell origin, encompassing a diverse array of clinical presentations.^{1–3} The clinical manifestations of LCH span from solitary lesions occurring in a singular organ, primarily the bone, to a more severe and widespread multisystemic disease.⁴ Two mechanisms have been proposed for the development of LCH: inflammatory and neoplastic mechanisms. There have been several reports of LCH subsequent to head trauma.^{2,4–12} However, rapid enlargement of the skull LCH within 1 month after a head injury is extremely rare. LCH preceded by head injury may have been caused by an inflammatory mechanism; however, to the best of our knowledge, there have been no reports demonstrating a local inflammatory response by histopathology. We present the rare case of a 3-year-old boy with right frontal

LCH that markedly enlarged 12 days after a repeat frontal head injury. We review the literature and discuss possible pathophysiology.

Illustrative Case

A 3-year-old boy incurred a right frontal impact head injury with no apparent neurological deficit. He subsequently bruised the same region multiple times. The right frontal swelling gradually enlarged over 6 days after the initial head injury. The frontal subcutaneous swelling had enlarged markedly as of 12 days after the initial head injury (Fig. 1A). Skull radiography revealed bone destruction at the same location. Computed tomography (CT) revealed more extensive destruction of the outer plate than the inner, with an isodensity mass in the diploe, and multiple subcutaneous cystic mass lesions (Fig. 1B and C). Follow-up CT 3 days

ABBREVIATIONS CT = computed tomography; EG = eosinophilic granuloma; IL = interleukin; LCH = Langerhans cell histiocytosis; MRI = magnetic resonance imaging; SS-LCH = single-system-Langerhans cell histiocytosis; TNF- α = tumor necrosis factor- α .

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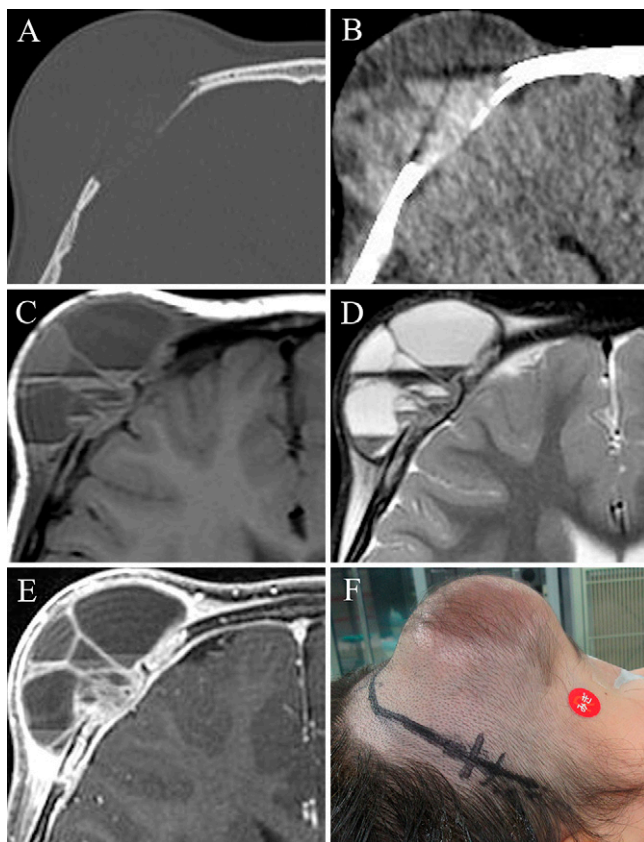


FIG. 1. A: Macroscopy shows the frontal subcutaneous swelling. **B and C:** CT scans with bone and brain windows 15 days after the initial head injury showing more extensive destruction to the outer plate than the inner, with an isodensity mass in the diploe, and multiple subcutaneous cystic mass lesions. **D:** T1-weighted MRI shows low-intensity alterations in the surrounding bone marrow. **E:** T2-weighted MRI shows an expansive osteolytic cystic mass exhibiting multiple fluid–fluid levels. **F:** Postcontrast T1-weighted MRI shows enhancement of the adjacent pericranial soft tissue, dura mater, and bone marrow and the mass itself.

later revealed further enlargement of the cystic lesions. Magnetic resonance imaging (MRI) revealed a uniform contrast effect at the frontal bony tumorous lesion (Fig. 1D–F). Cystic lesions were accompanied by a component of blood. A CT of the cervical to pelvic region revealed no neoplastic lesions. Furthermore, technetium-99 m methylene diphosphate scintigraphy showed no abnormal accumulation, except in the right frontal bone. The patient underwent open biopsy of the skull lesion and evacuation of the subcutaneous lesion. Histopathological examination with hematoxylin and eosin staining confirmed medium-sized cells with pale eosinophilic cytoplasm and grooved nuclei and an inflammatory cell infiltrate that included neutrophils, eosinophils, and lymphocytes (Fig. 2). Immunostaining confirmed the diagnosis of LCH with immunopositivity for CD1a and langerin (Fig. 3A and B). Immunohistochemical evaluation revealed positivity for interleukin (IL)-6 and tumor necrosis factor- α (TNF- α) and no immunopositivity for *BRAF* V600E (Fig. 3C–E). Inflammatory reactions after a head injury can induce rapid enlargement of LCH, whereas an oncogenic reaction via the *BRAF* gene may not manifest this symptom. The subcutaneous lesion reduced in size and did not recur, and it spontaneously and gradually disappeared over the course of 30 days after the biopsy (Fig. 4A and B). CT revealed improvement of the bony

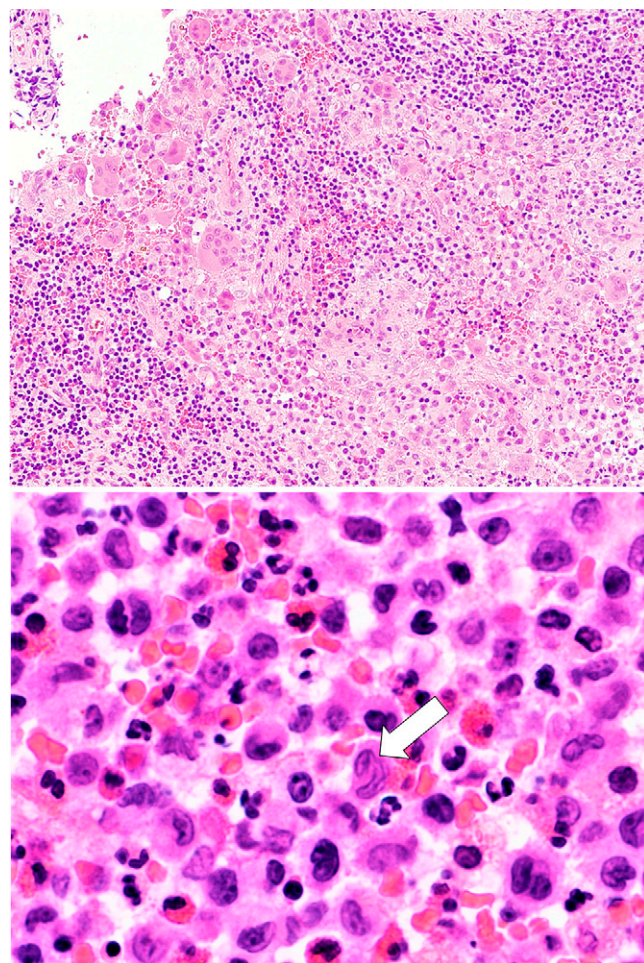


FIG. 2. Histopathological examination at original magnification $\times 100$ (upper) and $\times 400$ (lower) with hematoxylin and eosin staining confirmed medium-sized cells with pale eosinophilic cytoplasm and grooved nucleus (arrow), and inflammatory cell infiltrate including neutrophils, eosinophils, and lymphocytes.

lesion at 40 days postoperatively (Fig. 4C). There has been no recurrence after spontaneous remission of the lesion 1.5 years postoperatively.

Patient Informed Consent

The necessary patient informed consent was obtained in this study.

Discussion

Observations

LCH encapsulates a multitude of varied clinical presentations. Singular involvement of the osseous structure is designated as an eosinophilic granuloma (EG), whereas systemic participation can be classified as either Hand-Schuller-Christian disease or the Letterer-Siwe syndrome.⁶ It is estimated that within the broader incidence of LCH (0.5 per 100,000 children annually in the United States), approximately 70% of cases are delineated as EG.⁶

The majority of pediatric patients with an EG involving the skull will exhibit an expanding cranial mass devoid of symptoms, commonly referred to as a “clinically silent mass.”^{13–15} Trauma preceding a diagnosis of LCH has been observed and reported.^{2,3,5} Rawlings et al.¹⁵ reported a preceding trauma in 38% of their cases. Other

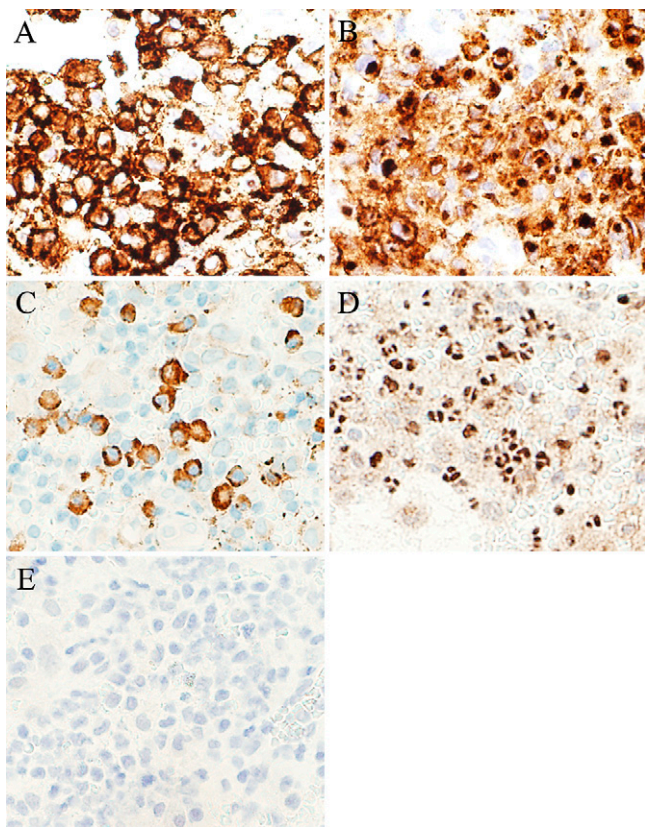


FIG. 3. **A:** Lesion stains positive for CD1a. **B:** Lesion stains positive for langerin. **C:** Lesion stains positive for IL-6. **D:** Lesion stains positive for TNF- α . **E:** Lesion stains negative for *BRAF V600E*. Original magnification $\times 400$ (A–E).

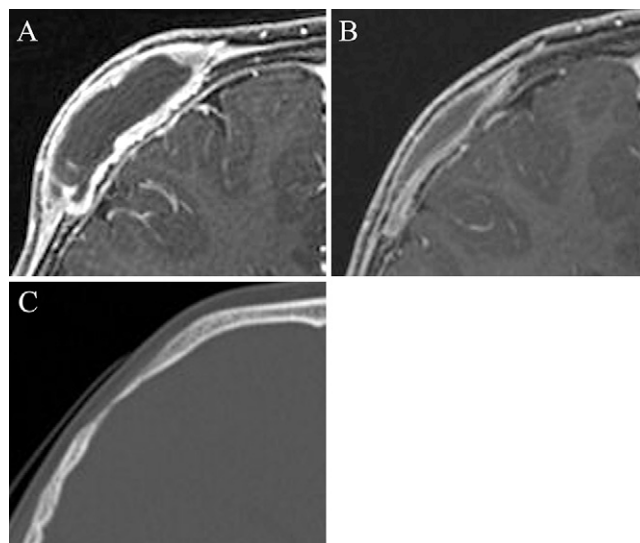


FIG. 4. **A:** Postcontrast T1-weighted MRI on postoperative day 7 shows size reduction. **B:** Postcontrast T1-weighted MRI on postoperative day 30 shows spontaneous remission and no recurrence. **C:** CT scan on postoperative day 40 shows improvement of a bony lesion.

studies have documented antecedent head trauma in 33%–50% of their cases.¹⁴ However, the pathophysiological mechanisms underlying the occurrence and enlargement of LCH after trauma remain uncertain.^{3,5} Some authors have postulated that trauma serves as a mere triggering event that draws attention to previously unnoticed EG lesions.^{4,5} Traumatic injury to LCH lesions may contribute to the etiology of extradural hematoma formation within these lesions.^{2,14} Head trauma has the potential to impact preexisting LCH lesions directly, resulting in secondary tumor rupture and subsequent hemorrhage.² Mechanical stimulation and biological inflammatory stimulation caused by head injury may induce inflammatory cascade reactions in an LCH lesion and cause marked, rapid growth of the lesion.

The pathophysiology of LCH remains obscure. However, the expansion of Langerhans cells could potentially be instigated by factors such as viral infection, immune dysfunction, aberrant cellular signaling proliferation, or bacterial infection.^{16,17} Enhanced genomic technology signifies a paradigm shift in the realm of LCH biology.¹⁶ Leveraging the OncoMap pyrosequencing platform, Badalian-Very et al.¹⁸ achieved a groundbreaking milestone by detecting the *BRAF V600E* mutation in an astounding 57% of LCH lesions. This seminal finding has since been corroborated by other studies, implicating its significance in the pathogenesis of LCH. The *BRAF V600E* mutation confers constitutive activation upon the MAPK pathway.¹⁹ Although somatic *BRAF V600E* mutations are encountered in a mere 7% of all human malignancies, their presence is also frequent in benign conditions such as melanocytic nevi and colon polyps.^{16,19}

The inflammatory process has also been reported to be involved in the mechanism of LCH. The clinical presentations of LCH, encompassing aggressive chronic granuloma formation, bone resorption, and soft tissue lesions, occasionally accompanied by neurodegeneration, resemble those observed in other IL-17A-related human disorders such as mycobacterial infection, Crohn's disease, rheumatoid arthritis, and multiple sclerosis. In a particular investigation, peripheral blood samples obtained from patients with LCH were subjected to cultivation, wherein the resultant monocyte-derived dendritic cells demonstrated an extended lifespan and a propensity for cell fusion, culminating in the formation of multinucleated giant cells.²⁰ These distinctive attributes position them as promising candidates for the genesis of characteristic granulomas. Oh et al.²¹ reported the quiescence of LCH with or without histological examination. They proposed the following clinical markers for assessing quiescence: absence of rapid progression within a fortnight, absence of severe pain or functional impairment, absence of fever, regression of skull manifestations attributed to a calvarial lesion, and biochemical evidence of inflammation, including elevated white blood cell count, erythrocyte sedimentation rate, and thrombocytosis.²¹

An inflammatory response has been suggested as one of the mechanisms of LCH, but to the best of our knowledge, there have been no reports demonstrating this histopathologically. This is the first study to report the presence of a local inflammatory response by immunostaining.

The spontaneous remission of LCH of the skull induced by repeat head trauma as reported here is rare. LCH induced by head trauma tends toward being found in the head, in boys, and without recurrence (Table 1). According to the LCH guidelines, the current classification distinguishes between single system disease (single-system-Langerhans cell histiocytosis [SS-LCH]) and multisystem disease, with the differentiation based on the extent of involvement at the time of diagnosis.²² Smaller lesions (<2 cm) may warrant complete excision for both diagnostic confirmation and therapeutic purposes. However, larger lesions (>5 cm) are not recommended for radical excision because of the potential increase in bony defect size, potential delay in healing, and risk of permanent skeletal defects. Lesions ranging from 2 to 5 cm in

TABLE 1. Summary for previously published case of traumatic calvarial LCH

Authors & Year	Age (yrs)/Sex	Lesion	Complication	Treatment	FU (mos)	Recurrence w/in FU
Lee et al., 2000 ²	8/M	Occipital	Epidural hematoma	Resection	NA	None
Martinez-Lage et al., 2002 ⁵	9/M	Occipital	Epidural hematoma	Resection	10	None
Lee et al., 2008 ⁴	10/M	Frontal	Epidural hematoma	Resection	NA	None
Pawar et al., 2011 ⁶	4/M	Rt parietal	Epidural hematoma	Resection	NA	None
Hollon et al., 2015 ⁷	2/M	Lt frontal	None	Resection	6	None
Karki et al., 2015 ¹¹	4/M	Lt temporal	None	Biopsy & steroid	5	None
	8/M	Rt parietal	None	Resection	84	None
Matsushita et al., 2020 ⁸	10/F	Lt parietal	None	Resection or biopsy	NA	NA
Mishra et al., 2021 ⁹	10/M	Rt temporo-parietal	None	Resection	1	None
Present case	3/M	Rt frontal	None	Biopsy	18	None

FU = follow-up; NA = not available

diameter are typically considered for biopsy and partial curettage.²² In cases in which the lesion affects the skull base, temporal bone, orbits, or vertebral column and involves adjacent soft tissues, systemic therapy is warranted to address the potential development of sequelae. Magaton et al.²³ documented a case involving a 45-year-old female patient who received a diagnosis of systemic LCH with multiple manifestations in the lung, liver, and bladder. Remarkably, after undergoing a single course of vinblastine-prednisolone therapy, all lesions completely resolved within a span of 2 months. The authors considered that smoking cessation caused the impressive remission. There are no previous reports of spontaneous remission after only aspiration for biopsy after marked aggravation. In the present case, the skull lesions surprisingly disappeared without specific treatment. Continuous inflammation caused by a repeat head injury may trigger the development of skull and subcutaneous lesions and accelerate abrogation. Evacuation of a subcutaneous lesion may terminate inflammatory cascade reactions caused by intracystic fluid and result in gradual remission. In the present case, *BRAF V600E* mutation was negative. Therefore, neoplastic cellular proliferation could not be the source of and entirely explain the spontaneous remission of the LCH. Extrinsic stimulation, such as mechanical injury or chemical injury, may contribute to the initiation of LCH and inflammatory progression, whereas the cessation of stimulation may have been the cause of the remission. The immunostaining analysis confirmed the presence of inflammatory cytokines, including IL-6 and TNF- α , thereby indicating the involvement of a localized inflammatory response in the pathogenesis of LCH.

The limitations of this report are twofold: first, it is not known whether the head injury preceded the onset of LCH, and second, although the presence of a local inflammatory response was demonstrated, similar findings may occur in LCH caused by a neoplastic mechanism such as the *BRAF V600E* mutation. It is necessary to accumulate cases to reveal the mechanism of LCH that develops by head injury.

Lessons

We treated a 3-year-old boy with right frontal LCH markedly enlarged 12 days after a repeat frontal head injury. Rapid enlargement of the skull LCH induced by a head injury is relatively rare. The pathophysiological mechanisms underlying the development of skull LCH after head injury remain unclear, and to the best of our knowledge, there is a lack of reports demonstrating the presence of a localized inflammatory response through immunostaining. Histopathological examination suggested the possibility

that the inflammatory reaction after the head injury could have been the cause of rapid enlargement of the LCH. The skull and subcutaneous lesions spontaneously disappeared without specific treatment. Our findings suggest that aggressive treatment is not always necessary in isolated bone lesions of SS-LCH because spontaneous resolution and bone healing may occur.

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Disclosures

Dr. Kudo reported a patent for Chimeric receptor that triggers antibody-dependent cell cytotoxicity against multiple tumors (13243N-PCT) with royalties paid.

Author Contributions

Conception and design: Saito, Ueno, Katayama, Mizukami, Nomura, Watanabe, Kinoshita, Asano, Kurose. Acquisition of data: Ueno, Watanabe, Sasaki, Fujiwara, Asano, Terui, Kurose. Analysis and interpretation of data: Watanabe, Terui, Kurose. Drafting the article: Saito, Ueno, Asano, Kurose. Critically revising the article: Saito, Ueno, Kudo, Kurose. Reviewed submitted version of manuscript: Kakuta, Morita, Kamio, Kudo, Terui, Kurose. Approved the final version of the manuscript on behalf of all authors: Saito. Statistical analysis: Kurose. Administrative/technical/material support: Asano, Kurose. Study supervision: Saito, Morita, Asano, Kurose. Clinical treatment strategy: Kudo.

Supplemental Information

Previous Presentations

This case has been presented at the 61st Annual Meeting of the Japan Neurosurgical Society Tohoku Branch, September 4, 2021, online only.

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